

Sentiment Analysis of Western vs. Non-Western Conflict News Coverage

Project Proposal

Artificial Intelligence | Summer 2026

Submitted: June 23, 2026

1. Problem Statement and Importance

New media doesn't cover conflicts in a vacuum. The words an outlet uses--which actors it foregrounds, whose deaths get counted, which side acts and which side suffers losses, etc--reflect where that outlet sits, politically and geographically. Most NLP work on this problem studies Western sources in isolation, which makes Western framing the implicit default rather than just one perspective among many. The actual question--whether outlets from different parts of the world cover the same event in measurably different tones--goes largely unasked. This project intends to ask it. The goal is to compare sentiment across Western outlets (BBC, Reuters, AP, New York Times) and non-Western outlets (Al Jazeera, RT, CGTN, Dawn, Tehran Times) covering the same conflict events, using five sentiment methods, and see whether the gap is real and, if so, how large it is.

2. Methods

Five methods are used, ranging from simple lexicon tools to a fine-tuned transformer. Running all five on the same corpus is important--if they agree, the finding is solid; where they disagree is itself informative as well.

- **VADER:** A rule-based lexicon that will score text from -1 to +1 using roughly 7,500 weighted features. It is fast, requires no labelled data, and is readable enough to sanity-check it by hand. This will be used as the baseline.
- **TextBlob:** This will add a subjectivity score from 0 to 1 alongside polarity, which is very important in this context--a neutral-sounding article can still be opinionated, and this distinction is worth tracking across outlets.
- **Naive Bayes:** It is a supervised classifier trained on labelled samples. This will be used to establish a learned baseline so that model-derived patterns can be compared against what the lexicon tools may pick up.
- **SVM:** Strong on high-dimensional text. Prior work on conflict coverage classification has hit 91%+ accuracy with bigram features, so it is a reasonable benchmark for this task.
- **BERT:** Fine-tuned on the labelled samples. It handles equivocated language and passive constructions that VADER and TextBlob will miss entirely. It is the most sensitive method, but also the most expensive to validate.

3. Experimental Setup

Data Collection

Articles will come from two sources: NewsAPI for full article text pulled using outlet and keyword, and GDELT for historical coverage and built-in tone scores that can serve as a cross-check on model outputs. Outlets will be manually labelled as Western or non-Western, with at least three per group. Articles are filtered to a single conflict even or tight event cluster so as to hold the topic constant across the comparison.

Labels

For *Naive Bayes* and *SVM*, labels will come from *VADER* scores used as weak supervision for the full corpus, with a manually verified sample size of about 200-300 articles on top. *BERT* will get fine-tuned on the labelled sample, and if time runs short then an off-the-shelf sentiment checkpoint will be used instead.

Evaluation Metrics

- Mean polarity per group (Western vs non-Western), tested with a t-test using bootstrap confidence intervals added to quantify the uncertainty around the effect size.
- Accuracy, Precision, Recall, and F1 for the supervised models on the held-out labelled set.
- Agreement between methods via Spearman correlation--disagreements between *VADER* and *BERT* are as interesting *as* agreements.
- TextBlob subjectivity distribution across groups, as a rough proxy for how opinionated each outlet's coverage is.

Expected Output

The main result is a comparison of sentiment scores across outlet groups for all five methods, showing whether and by how much Western and non-Western coverages diverges on the same events. A secondary finding will be which method is most sensitive to the distinction, and where the simpler tools may fall short.